

ECE330 – Fall 2024
Exam 1
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90 min

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NAME Solution

NetID _____

Instructions:

- Clearly PRINT your name in CAPITAL LETTERS.
- Clearly write your NetID.
- This exam is closed book and closed notes.
- To get credit, you must SHOW ALL your work and provide explanations where needed.
- You may write on both sides of the exam paper. If there is anything important on the reverse side, please indicate this on the front side.
- All answers must INCLUDE UNITS whenever appropriate.
- Graphing and/or 4-function calculators are permitted, but no cellphones/laptops.
- This exam booklet contains 9 pages (including this one) with 4 problems.

Problem 1 (25 Points)

A single-phase source is serving three loads connected in parallel:

- Load 1 draws 100 A at a lagging PF of 0.8.

- Load 2 draws 8 kW at a lagging PF of 0.7.

- Load 3 is an impedance of $(5+j2) \Omega$. $= 5.39 \angle 21.8^\circ$

The voltage across the loads is $v(t)_{load} = 283 \sin(377t)$ V. $= \frac{283}{\sqrt{2}} \angle -90^\circ$

The impedance of each of the two wires that serves this combination of loads is $(0.1+j0.1) \Omega$.

$$= (.141 \angle 45^\circ) \times 2$$

a) Find the total average and reactive powers provided by the source.

$$P = \underline{38.3 \text{ kW}}$$

$$Q = \underline{30.3 \text{ kVAR}}$$

Need I_{tot}

$$\text{Load 1: } |I| = 100 \text{ A, } \phi_v - \phi_i = \cos^{-1}(0.8) = 36.87^\circ$$

$$I = 100 \angle -126.87^\circ$$

$$\text{Load 2: } |I| = \frac{P}{PF \cdot V/\sqrt{2}} = 57.11 \quad \phi_v - \phi_i = \cos^{-1}(0.7) = 45.57^\circ$$

$$I = 57.11 \angle -135.57^\circ \quad \angle I = -90 - 45.54$$

$$\text{Load 3: } \frac{V_{load}}{\sqrt{2}} = ZI \quad I = \frac{V_{load}}{\sqrt{2} Z} = 37.13 \angle -111.8^\circ$$

$$\underline{I_{tot}} = I_1 + I_2 + I_3 = 192.3 \angle -126.57^\circ$$

$$V_{line} = I_{tot} Z_{line} = 54.2 \angle -81.57^\circ$$

$$\underline{V_{tot}} = V_{line} + V_{load} = 253.9 \angle -88.21^\circ$$

$$S_{tot} = V_{tot} I_{tot}^* = 48815 \angle 38.36^\circ$$

$$P = |S| \cos(\theta) = 38275 \text{ W}$$

$$Q = |S| \sin(\theta) = 30297 \text{ VAR}$$

b) Find the source voltage and current as time domain quantities. (from $\uparrow$)

$$i(t)_s = \underline{271.95 \cos(377t - 126.57^\circ) \text{ A}}$$

$$v(t)_s = \underline{359.1 \cos(377t - 88.21^\circ) \text{ V}}$$

$$I_m = \sqrt{2} I_{\text{RMS}}$$

$$V_m = \sqrt{2} V_{\text{RMS}}$$

c) What is the combined power factor over the three loads? State whether it is leading or lagging.

$$\text{Combined PF} = \underline{.803}$$

Circle one: Leading / Lagging

$$S_{\text{LOAD}} = V_{\text{LOAD}} I_{\text{tot}}^* = 38481 \angle 36.57$$

$$\cos(36.57) = .803 \text{ PF}$$

Problem 2 (25 Points)

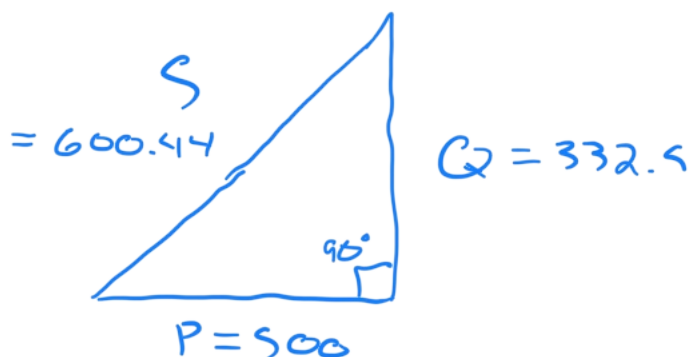
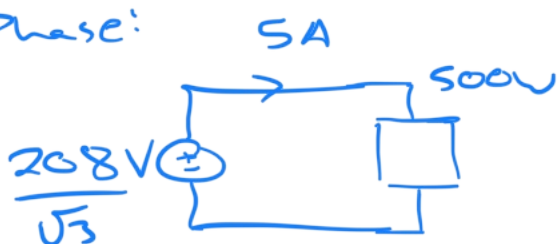
A 3-phase, 208 V (line-line), Wye-connected 60 Hz source supplies 5A (magnitude) of line current to a 1500 W, 3-phase wye-connected load at a lagging power factor.

- a) When a new wye-connected purely reactive load is connected in parallel with the 1500 W 3-phase load, the line current (magnitude) increases to 7A. What is the impedance per phase of the new wye-connected load?

Hint: You may assume that the real power does not change when the new wye-connected load is added.

$$\bar{Z}_Y = \underline{42 \angle 90 \Omega}$$

Per Phase:



old:

$$|S| = VI = 600.44$$

$$Q = \sqrt{S^2 - P^2} = 332.5$$

new: $|S|_{\text{new}} = IV = 840.6 \text{ VA}$



$$Q = 332.5 + Q_{\text{new}}$$

$$Q_{\text{tot, new}} = \sqrt{S^2 - P^2} = 675.8 = Q_{\text{old}} + Q_{\text{new}}$$

$$Q_{\text{new}} = 343.3 \text{ VAR} \Rightarrow \bar{S} = 343.3 \angle 90$$

$$I = \left(\frac{S}{V} \right)^* = 2.86 \angle -90$$

$$Z = \frac{V}{I} = 42 \angle 90$$

b) Now a delta-connected capacitor bank is added in parallel with the two loads, reducing the total line current from 7A to 4.75 A (magnitude). What is the size of the capacitors added per phase in Farads (capacitance) in the delta-connected capacitor bank?

$$C = \underline{24.6 \mu F}$$

$$|S_{\text{new}}| = \sqrt{3} V_L I_L = 1711.27 \text{ VAR}$$

$$Q_{\text{new}} = \sqrt{\left(\frac{S_{\text{new}}}{3}\right)^2 - P^2} = 274.6$$

$$Q_{\text{added}} = Q_{\text{new}} - Q_{\text{old}} = -401.2$$

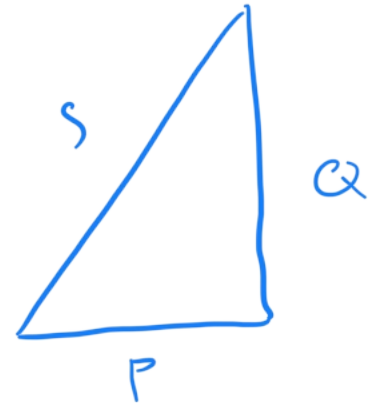
$$S = 401.3 \angle -90$$

$$I = \left(\frac{S}{V}\right)^* = 1.93 \angle 90$$

$$Z_{\Delta} = \frac{V}{I} = 108 \angle -90 = Z_C$$

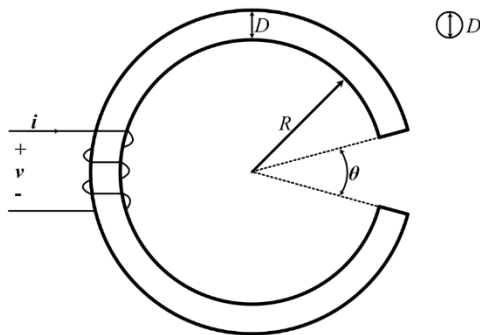
$$Z_C = \frac{-j}{2\pi\omega C} = \frac{1}{2\pi\omega C} \angle -90^\circ \quad \omega = 60$$

$$C = 2.46 \text{ E-5 F}$$



Problem 3 (25 Points)

A toroidal iron core with permeability $\mu = 1500\mu_0$ and inner radius $R=10\text{cm}$, shown below, has a coil with 100 turns wrapped around it. The cross section of the toroid is a circle, as shown, and has a diameter $D=2\text{ cm}$. There is a cut in the toroid that spans an angle $\theta=10^\circ$. Fringing effects can be neglected.



a) What are the reluctances for the iron core and the air gap?

Hint: the arc length of an arc lying on a circle with radius r spanning θ radians is $l = r\theta$.

$$R_{\text{core}} = \underline{1,134,797 \frac{\text{Ampere-turns}}{\text{Weber}}}$$

$$R_{\text{gap}} = \underline{48,655,343 \frac{\text{Ampere-turns}}{\text{Weber}}}$$

$$R = \frac{l}{\mu A}$$

$$\text{core: } l = 2\pi(11\text{cm}) \frac{35}{36} = .672\text{m}$$

$$\mu = \mu_0 \times 1500$$

$$A = \pi(1\text{cm})^2 = 3.14\text{E-4}$$

$$\text{air gap: } l = 2\pi(11\text{cm}) \frac{1}{36} = .019\text{m}$$

$$\mu = \mu_0$$

$$A = \pi(1\text{cm})^2 = 3.14\text{E-4}$$

b) What is the total magnetic flux?

$\phi =$ _____

$$\phi = \frac{Ni}{R_{tot}} = \frac{100i}{R_{core} + R_{gap}} \text{ Wb}$$

c) What is the inductance of this system?

$L =$ 2.008E-4 H

$$L = \frac{N\phi}{i} = \frac{N^2}{R_{Tot}} = 2.008E-4$$

Problem 4 (25 Points)

Note: Parts **a)** and **b)** are unrelated.

- a)** An ideal 240V/120V, 60 Hz single-phase transformer is being used to supply power to a load drawing 1kW at 0.8 PF lagging. Note that 240V corresponds to the primary side, and 120V corresponds to the secondary side.

- i. Find the complex power being supplied to the load.

$$\bar{S}_L = \underline{1250 \angle 36.9 \text{ VA}}$$

$$|S| = \frac{P}{PF} = 1250 \text{ VA} \quad \angle S = \cos^{-1}(0.8) = 36.9$$

- ii. Find the load impedance.

$$\bar{Z}_L = \underline{11.52 \angle 36.9 \Omega}$$

$$I = \left(\frac{S}{V}\right)^* = 10.42 \angle -36.9$$

$$Z = \frac{V}{I} = 11.52 \angle 36.9 \Omega$$

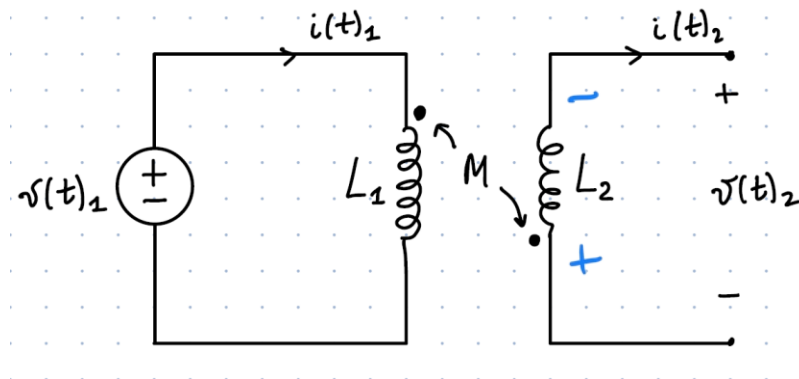
- iii. Find the load impedance as seen from the primary side.

$$\bar{Z}_p = \underline{46.07 \angle 36.9 \Omega}$$

$$Z_p = a^2 Z_s \quad a = \frac{N_1}{N_2} = 2$$

$$Z_p = 4 Z_s$$

b) Consider two coils magnetically coupled together as shown on the diagram below.



Self-inductances of the coils are given as $L_1 = 2 \text{ H}$ and $L_2 = 250 \text{ H}$. Mutual inductance between two coils is given as $M = 5 \text{ H}$. Coil 2 is open-circuited. What is the voltage $v(t)_2$ when a sinusoidal voltage $v(t)_1 = 1\sqrt{2} \cos(377t) \text{ V}$ is applied to coil 1?

$$v(t)_2 = \underline{\underline{\frac{5\sqrt{2}}{2} \cos(377t - 180^\circ) \text{ V}}}$$

due to dot convention

$$-V_2 = M \frac{\partial i_1}{\partial t} + L_2 \frac{\partial i_2}{\partial t}$$

$$V_1 = L_1 \frac{\partial i_1}{\partial t} + M \frac{\partial i_2}{\partial t}$$

$$\frac{\partial i_2}{\partial t} = 0 \rightarrow \text{open circuit}$$

$$V_1 = L \frac{\partial i_1}{\partial t} \Rightarrow \frac{\partial i_1}{\partial t} = \frac{V_1}{L} = \frac{\sqrt{2}}{2} \cos(377t)$$

$$-V_2 = \frac{\partial i_1}{\partial t} M = -\frac{5\sqrt{2}}{2} \cos(377t)$$

$$= \frac{5\sqrt{2}}{2} \cos(377t - 180^\circ)$$